

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior backend engineer specialized in auth flows, RBAC, and event metering pipelines across Node.js + TypeScript + PostgreSQL. Led the Parse → Firebase migration of 20+ apps at MTailor (zero downtime, \$5K/month AWS savings), built an idempotent [Pub/Sub event pipeline](#) processing millions of events, and shipped customer-facing revenue features (Express Shipping, +\$40/order). Now applying the same operational discipline to IAM and usage-based billing infrastructure at LLMOps scale.

CORE COMPETENCIES

TypeScript / Node.js (5+ yrs)

PostgreSQL & SQL

Authentication & RBAC

Usage-based billing & metering

Event pipelines (Pub/Sub, Kafka)

Idempotent consumers

Distributed systems

Zero-downtime migrations

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer (US Remote, reports to CTO)

- Led end-to-end migration of 20+ production apps from Parse/MongoDB to Firebase/Firestore, zero downtime, reporting to the CTO.
- Built **real-time event metering pipeline** (MongoDB ↔ Firestore via GCP Pub/Sub) with idempotent consumers and dead-letter queue — same pattern needed for usage-based billing at scale.
- Designed customer-facing 3D visualization (ffmpeg + video overlay) increasing buyer conversion on product detail pages.
- Migrated file storage from AWS S3 to GCS via a streaming Python pipeline with integrity checks across 20+ apps.
- Saved the company **\$5,000/month** by re-architecting services off AWS onto Firebase/GCP — designed cutover plan, ran shadow reads, executed without downtime.
- Implemented Express Shipping revenue feature end-to-end — backend pricing pipeline, UI, analytics — generating **+\$40 per order**.

Stack: TypeScript, Node.js, Firebase, GCP Pub/Sub, AWS, Python, ffmpeg, React

MTailor (Contractor)

Mar 2022 – Jul 2022

Backend Migration Contractor (US Remote)

- Hired via Upwork to execute backend migration from Parse/MongoDB to Firebase/GCP.
- Built migration script reading MongoDB and writing to Firestore with complex transformation logic.
- Documented new Firebase SDKs (Node + Python) and trained Ops team on the new dashboard + backup procedures.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer (Singapore Remote)

- Built batch query layer reading **2M+ Firestore records** with paginated cursors + pre-aggregated views — kept Morningstar prices fresh in the PWA at low cost.
- Extended backend APIs using Firebase Cloud Functions + Express.js on Heroku.
- Shipped responsive UI pages with HTML, Tailwind, React.

Mind2Matter

Jan 2022 – Jul 2022

Software Engineer (US Remote, reports to CTO)

- Built backends for DeFi applications on Node.js + Web3.
- Delivered React UIs from Figma under tight timelines.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Tech Lead (Uganda Remote)

- Led a team of 4 building legal-tech systems for NGOs (FIDA Uganda, LASPNET Uganda).
- Architected a **microservices backend** with **Kafka + Docker + Kubernetes** for FIDA's case management app — service mesh with gRPC + Protobufs.
- Built USSD service for legal aid bridging to Node.js backend via gRPC.
- Built cross-platform mobile app (React Native + Expo, iOS + Android) with Firebase push.

PROJECTS

Parse → Firebase Migration (MTailor) zero downtime

Migrated 20+ production apps from Parse/MongoDB to Firebase/Firestore with dual-write Pub/Sub pipeline, per-app flag flip, and rollback discipline. Saved \$5K/month in AWS spend with no customer-visible incident.

Node.js, Pub/Sub, Firestore, MongoDB, GCP, AWS, Python

Healthcare WhatsApp Chatbot OpenAI + Vision

Built prescription-OCR chatbot on WhatsApp Business API integrating **OpenAI** + **Google Vision OCR**. Low-latency request flow with retry + observability around third-party LLM calls.

Node.js, OpenAI, Google Vision, WhatsApp Business API, GCP

Real-time Pub/Sub Sync Pipeline millions of events

Designed idempotent Pub/Sub consumer with dead-letter queue + replay for MongoDB ↔ Firestore sync. Pipeline ran parallel production traffic during 6-month migration with zero data loss.

Node.js, GCP Pub/Sub, Firestore, MongoDB, idempotency keys

Marketplace Recommendation Engine

Built collaborative + content-based filtering engine in MongoDB for service-provider matching. Ranked candidates from interaction history + skill graphs.

Node.js, MongoDB, vector similarity

EDUCATION

B.Sc. Computer Engineering — Makerere University

Aug 2018 – Sep 2022

Kampala, Uganda

CERTIFICATIONS

Open-source contributor & GitHub portfolio — github.com/zac-09

Ongoing

SKILLS

Languages: TypeScript, Node.js, JavaScript, Python, Go (intermediate), SQL

Backend / Infra: PostgreSQL, MongoDB, Firestore, GCP Pub/Sub, Kafka, gRPC, Docker, Kubernetes, AWS, GCP

Domain: RBAC, API key auth, event metering, idempotent pipelines, OAuth/SSO (ramping)

Tools: Git, Linux, OpenAI APIs, Google Vision